Model Question Paper-1/2 with effect from 2023-24

USN | | | | | | | | | | |

Fourth Semester B.E. Degree Examination

Subject Title: Microcontrollers

TIME: 03 Hours **Max. Marks: 100**

Note: 01. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**.

		Module -1	***Bloom's Taxonomy Level**	**Marks**
Q.01	a	Mention the difference between 1. Microctroller and Microprocessor 2. RISC and CISC	L2	10 Marks
	b	Explain the architecture of an ARM embedded device with the help of neat diagram.	L2	10 Marks
		OR		
Q.02	a	Explain in detail about Current Program Status Register (CPSR).	L2	10 Marks
	b	With a neat diagram, explain embedded system Hardware.	L2	10 Marks
		Module-2		
Q. 03	a	Explain different arithmetic instructions in ARM processor with an example.	L2	10 Marks
	b	Explain single register load store addressing mode syntax, table, index mode with an example.	L2	10 Marks
		OR		
Q.04	a	Explain barrel shifter instructions in ARM with suitable examples.	L2	10 Marks
	b	Explain different Logical instructions in ARM processor with an example.	L2	10 Marks
		Module-3		
Q. 05	a	Explain code optimization, profiling and cycle counting.	L3	10 Marks
	b	Write a C program that prints the square of the integers between 0 to 9 using functions and explain how to convert this C function to an assembly function with command.	L3	10 Marks
		OR		
Q. 06	a	Discuss how registers are allocated to optimize the program.	L3	10 Marks
	b	Develop an ALP to find the sum of first 10 integer numbers.	L3	10 Marks
		Module-4		
Q. 07	a	With a neat diagram explain ARM processor exceptions and modes.	L2	10 Marks
	b	Explain assigning interrupts and interrupt latency.	L2	10 Marks
		OR		
Q. 08	a	Briefly explain what happens when an IRQ and FIQ exception is raised with an ARM processor.	L2	10 Marks

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	b	Explain firmware execution flow and explain Red Hat RedBoot.	L2	10 Marks
Module-5				
Q. 09	a	Explain the basic architecture of cache memory.	L2	10 Marks
	b	Explain how main memory maps to a cache memory.	L2	10 Marks
OR				
Q. 10	a	With a neat block diagram explain associative cache.	L2	10 Marks
	b	Briefly explain cache line replacement policies.	L2	10 Marks

*Bloom's Taxonomy Level: Indicate as L1, L2, L3, L4, etc. It is also desirable to indicate the COs and POs to be attained by every bit of questions.

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